

2^k Factorial Design

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Introduction

In this report we are going to apply the 2^k FD method for the following data:

```
#Response vector
Y<-c(28,25,27,36,32,32,18,19,23,31,30,29)

#The factors
B <- rep(c(-1, 1), each = 6)      # repeats each level for half the total
A <- rep(rep(c(-1, 1), each = 3), 2) # repeats each level 3 times
AB <- A*B

#We create the design matrix
Design <- data.frame(
  Intercept = 1,
  A = A,
  B = B,
  AB = AB
)
Design
```

	Intercept	A	B	AB
1	1	-1	-1	1
2	1	-1	-1	1
3	1	-1	-1	1
4	1	1	-1	-1
5	1	1	-1	-1
6	1	1	-1	-1
7	1	-1	1	-1
8	1	-1	1	-1
9	1	-1	1	-1

```

10      1  1  1  1
11      1  1  1  1
12      1  1  1  1

```

We then calculate the estimator:

```

X <- cbind(1, A, B, AB)
beta_hat <- solve(t(X) %*% X) %*% t(X) %*% Y
beta_hat

```

```

      [,1]
27.5000000
A   4.1666667
B  -2.5000000
AB  0.8333333

```

We see that the analysis matches the results we expected and the results we derived by doing the calculations by hand. We will now do it with a single line of code to calculate p-values as well.

```

df <- data.frame(Y, A, B)
model <- lm(Y ~ A * B, data = df)
summary(model)

```

Call:

```
lm(formula = Y ~ A * B, data = df)
```

Residuals:

```

      Min       1Q   Median       3Q      Max
-2.000 -1.333 -0.500  1.083  3.000

```

Coefficients:

```

              Estimate Std. Error t value Pr(>|t|)
(Intercept)  27.5000    0.5713   48.135 3.84e-11 ***
A              4.1667    0.5713    7.293 8.44e-05 ***
B            -2.5000    0.5713   -4.376 0.00236 **
A:B             0.8333    0.5713    1.459 0.18278
---

```

```

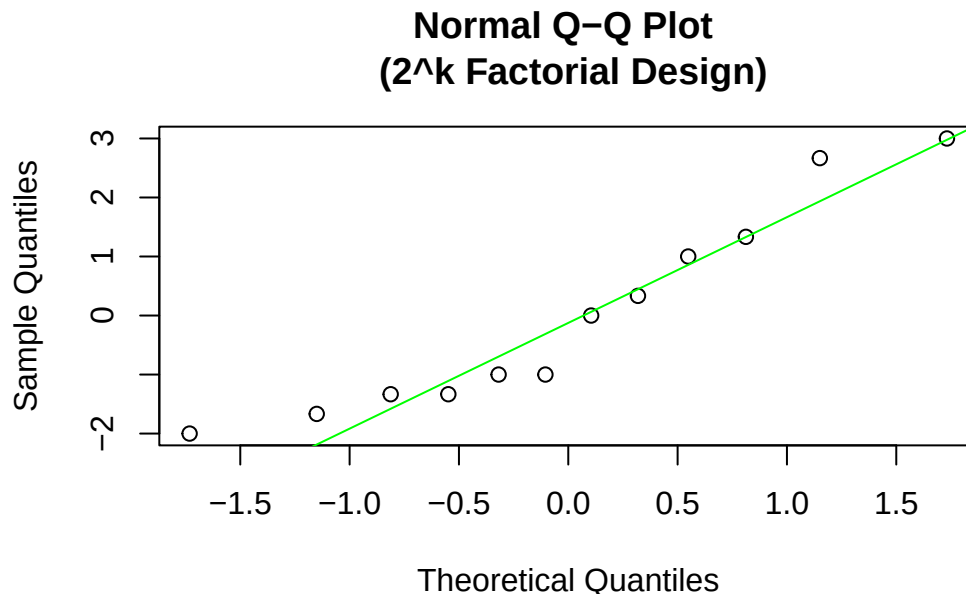
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Residual standard error: 1.979 on 8 degrees of freedom
Multiple R-squared: 0.903, Adjusted R-squared: 0.8666
F-statistic: 24.82 on 3 and 8 DF, p-value: 0.0002093

Of course no analysis would be complete without a Normal Probability Plot:

```
qqnorm(residuals(model),main = "Normal Q-Q Plot \n (2^k Factorial Design)")  
qqline(residuals(model),col="green")
```



We see the residuals roughly fall along a line. The correlation is not great but we can see that the errors follow a normal distribution.